Handwritten signature forgery detection using Deep Neural Network

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Abstract— **A common biometric for establishing a person's identity in document forensics is verification of handwritten signatures. Signature plays an important role in banking, financial, commercial and so on. However, issues associated with signatures are numerous since any two signatures may look very similar with little to no differences written by the same person. Despite the enormous research efforts, offline signature verification is still difficult, especially when attempting to differentiate between competent forgeries and authentic signatures because the visual differences between the two can sometimes be less noticeable than between two authentic ones. Due to the expanding usage of digital signatures and electronic documents, there has been an increase in the demand for effective and precise signature forgery detection systems (SFDS). In order to effectively combat handwritten signature forgery, this study intends to create an automated system utilizing a deep neural network (DNN) especially a simple and efficient Convolutional Neural Network (CNN) that can determine if a given signature is real or faked. The model will take signature image as input and extract important features through different layers of CNN that will helpful in distinguishing genuine and forged signatures. The model performance is also accessed with different image resizing techniques and optimizers used in training of CNN. The model's performance in terms of a set of assessment metrics, such as accuracy, precision, recall, and F1-score are provided in this paper after extensive tests are conducted across a huge dataset of handwritten signatures including both genuine and various types of forged signatures. Additionally, the model's performance is also contrasted with a cutting-edge deep learning model. The simulation results conclusively demonstrate the appropriateness of the suggested methodology for automatically detecting forgery in handwritten signature images.**

Keywords— Forgery detection, Classification, Deeplearning, CNN

I. INTRODUCTION

A signature is the way a person writes their own name in a special and consistent manner. It is a unique and personal expression of one's identity. A common biometric for establishing a person's identity in document forensics is verification of handwritten signatures. Signature serves as a form of biometric authentication or verification of the person's identity or consent on various kinds of documents that require such confirmation. For example, a signature can be used to authorize a payment on a bank cheque, to agree to the terms and conditions of a contract, to indicate the sender or recipient of a letter, or to provide or request information on a form. However, issues associated with signatures are numerous since any two signatures may look very similar with little to no differences written by the same person. Traditional methods of signature verification rely on human experts or simple features extracted from the signature images. However, these methods are prone to errors and limitations. Despite the enormous research efforts, offline signature verification is still difficult, especially when attempting to differentiate between competent forgeries and authentic signatures because the visual differences between the two can sometimes be less noticeable than between two authentic ones [1]. The advancement of technologies in today's generation has also led to an amplified amount of fraudulent activities. One such type of fraud is a handwritten signature forgery, where in a fraud tries to imitate a person's signature to gain access to bank accounts and /or carry out various criminal activities. To mitigate such deceptive activities, and to prevent fraudulent authorization, there has been an increase in the demand for effective and precise signature forgery detection system (SFDS) that distinguishes between forged and genuine handwritten signatures.

In the literature, a number of SFDS models based on machine learning and deep learning have been laid out [1-10]. In [2] the author has proposed a method for offline signature recognition and forgery detection using a CNN model. The authors proposed a Crest-Trough algorithm with Harris-Surf for the forgery detection. Pre-processing techniques that include noise removal, scaling of the images, centralization and rotation of the signature images are used in the study. The proposed CNN model architecture is based upon 5 layers that use a back-propagation method for training the model. The model is tested on the CEDAR dataset that consists of 1320 images for each genuine and forged signature. Finally evaluation metrics such as False Acceptance Rate (FAR) and False Rejection Rate (FRR) is used judge to performance of the described CNN model.

A region based DNN using Mutual Signature DenseNet (MSDN) is explored in [3]. In order to learn the feature representation of handwritten signatures, it comprises of two CNN branches that share the same structure and parameters. In their proposed method the similarity between two signatures obtained by analysing multiple local regions by using the MSDN feature extractor is then fed forward to a Siamese Neural Network for training. The CEDAR dataset of handwritten signatures along with GPDS dataset is used for training the Neural Network. The model in then compared with different CNN architectures such as VGG-16, AlexNet, ResNet-18 and finally performance metrics

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such as FAR and FRR along with accuracy metrics are taken into account for model evaluation.

In [4] the authors have proposed the use of a Siamese Neural Networks to assist in resolving the offline handwritten signature verification problem in a writer-independent context. They include the image binarization in the pre-processing step to convert the original image to a single channel grey scale image. Then Morphological erosion is used to increase the signature strokes, with an image resizing of 128 x 128 pixels. Augmentation and synthetic signature generation is also performed. Performance of the model is then compared with other pre-trained networks such as LeNet and SigNet with accuracy and error rate metrics.

A simple CNN with three convolution and maxpooling layer arranged in alternate fashion with one flatten and dense layer is proposed in [5] for signature forgery detection. The model is tested on a dataset including 6000 images prepared by the author with one person making originals and two others making the forgeries. As all images are in RGB format, so they are converted to grey scale image in pre-processing stage. The greyscale images of the handwritten signatures are then converted to Bitmap format to rasterize the signature dataset. Accuracy and loss percentage metrics is used for model assessment.

In [6] another CNN model derived from the per-trained ResNet-50 as its base model, and adding additional layers on top is proposed as a SFDS. The model has used various autoencoders for feature extraction. The model is tested with a dataset including 2660 english signatures collected from kaggle dataset.

In [7] authors have used a CNN model for detecting signature forgery. The CNN model was trained and tested for various different dataset split ratios. Loss and accuracy was used to judge the performance of the model.

In [8] the authors have used attendance sheets to generate handwritten signature dataset. OMR (Optical Mark Recognition) tools were used to scan the attendance sheets, the OMR tools perform operations like correcting the orientation and identify its class by its unique barcode present in the attendance sheet. To distinguish between signatures and non-signature, a feed forward neural network was utilized. A deep CNN classifier was then proposed to check the legitimacy of the signatures. Various hyper-parameter tuning was also performed to increase the accuracy of the model. Optimizer like ADAM and SGD were also compared. A confusion matrix was used to evaluate the performance of the neural network model.

A comparison of a custom CNN, VGG16, a Siamese Neural Network(SNN) and a SNN with SigNet architecture is presented in [9] for signature forgery detection. By testing the model on CEDAR dataset, the custom SNN shows better accuracy compared to the other models.

A 14 layer CNN using 6 convolutions, 5 maxpooling, 1 flatten and 2 dense layers is proposed in [10] to verify authenticity of handwritten signatures. The model is tested on a dataset including 300 images with equal proportion of real and forged ones related to signatures of 30 people. The images are re-scaled to a standard image size of 512 x 512 pixels. The model performance is also accesses with three different split ratios, out of which the 80:20 ratio has shown the best accuracy in training and validation set.

Though numerous CNN architectures are proposed in literature, but the goal of developing a simple and more efficient one always attracts researcher in this field. In order to effectively combat handwritten signature forgery, this study intends to create an automated system utilizing a simple custom CNN that can determine if a given signature is real or faked. The model will take signature image as input and extract important features through different layers of CNN that will help in analyse and distinguish between genuine and forged signatures. The model performance is also accessed with different image resizing techniques and optimizers used in training of CNN. The model's performance in terms of a set of assessment metrics, such as accuracy, precision, recall, and F1-score are provided in this study after extensive tests are conducted across a huge dataset of handwritten signatures including both genuine and various types of forged signatures. Additionally, the model's performance is contrasted with a pre-trained VGG16 and a trainable VGG16. The simulation results conclusively demonstrate the appropriateness of the suggested methodology for automatically detecting forgery in handwritten signature images. The main contribution of the work is delineated as follows:

- A custom 7 layer CNN based SFDS is developed.
- The CNN's performance with 3 optimizers and 3 different training testing split ratios is analyzed.
- The performance of custom CNN is compared with the pre-trained and trainable VGG-16 model.

The next section details the proposed CNN based signature forgery detection model followed by the findings of the model in section 3 and conclusion in section 4.

II. Proposed CNN based SFDS

Signature forgery is a serious problem that can cause financial and legal damages to individuals and organizations. In this study, we propose a simple custom CNN model for signature forgery detection that can automatically learn discriminative features from the signature images and classify them as genuine or forged one, assuming it as a binary classification problem.

Fig. 1 depicts the layout of the developed 7-layer CNN based SFDS model including 3 convolution layers with associated maxpooling and batch normalization, one dropout layer, one flatten layer and two fully connected dense layers. The several convolutional, pooling, and fully connected layers included in the network helps to capture the subtle variations and characteristics of different signatures. The input layer, the convolution layer, the pooling layer, the flatten layer, the fully linked dense layer and the output layer are the six fundamental layers that make up the network. The signature images passing through the input layer are converted to matrix of pixels. The input layer's size is 227 x 227 x 3 pixels. In the first convolution layer, 32filters with kernel size of 3x3 has been used. The second convolution layers consist of 64 filters with 3x3 as the kernel size. The third convolution layers uses 128 filters with the same kernel size as the previous layer. All the three convolutional layers use rectified linear (ReLu) as the activation function. The first two convolution layers use MaxPooling while the third convolution layer uses GlobalAveragePooling followed by a fully connected dense layer with sigmoid activation function. Batch normalization is also performed after the

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first dense layer. Next a flatten layer is used to convert the resultant output from the previous layer to a single dimension. The flatten layer is then followed by a dropout layer with 20% dropout mechanism to overcome the over-fitting problem. Finally, one dense layer with the softmax activation function is used as the output layer.

Fig. 2 shows the overall principle of how the CNN based SFDS works. In the data acquisition stage, handwritten signatures were collected from the publicly available CEDAR dataset https://cedar.buffalo.edu/NIJ/data. This dataset includes a total of 2640 grey scale handwritten signatures out of which 1320 are genuine and 1320 are forged signatures. Further a sample of 60 images of each genuine and forged signature prepared by us is also added in the dataset making a total of 2760 images. To achieve uniformity among the images, which varied in size from 200px to 800px, we used the resize function of OpenCV (a powerful library for image processing) with INTER_LINEAR as the interpolation parameter that performs a linear interpolation of the pixel values while resizing the signature images. Then a Gaussian blur filter is applied to perform the noise removal from the images. The filter is applied using fastNlMeansDenoising function of the OpenCV image processing library with tuned values for Gaussian blur as the function parameters. After the pre-processing stage the dataset was split into training and testing sets with the specified splitting ratio. Then the proposed CNN model is created and then trained using different optimizers and 'binary_crossentropy' loss function, for addressing the binary classification task. Finally the derived model is accessed using accuracy and loss metrics to see how well the model has performed.

III. Experimaental Result Discussion

The proposed 7-layer CNN based SFDS was implemented using Python on a computer system configured with AMD Ryzen 5 3600 CPU at 4.2Ghz, Nvidia RTX 2060 6GB GPU(Graphics Processing Unit) and 16GB of 3200mhz RAM. Python version 3.9.5 was used in the study. The OpenCV image processing library was used to resize and apply filtering to the dataset. Tensorflow version 2.10 was used to create the sequential CNN model. Matplotlib was used to plot performance graphs for training and validation accuracy and loss. The SK.learn library provided us with functions to split the dataset, to show various metrics such as f1-score, recall and precision and to visualize the confusion matrix.

In this study, the performance of the proposed CNN based SFDS is observed with 6 different image resize. A randomly chosen handwritten signature image being resized in six different resolutions and filtered in pre-processing stage is depicted in **Fig. 3** Following pre-processing, the dataset was tested as shown in Table 1 by dividing it into training and test sets with different splitting ratios**.** The CNN model was experimanted with three popular optimizers such as ADAM, RMSprop and SGD and three different split ratios for 100 epochs. Each epoch used a batch size of 30. Learning rate for RMSprop and SGD optimizer was set to 0.001. Analysing the values of Table 1, it is evident that the SGD optimizer with the validation split set to 0.2 (80:20 split ratio) clearly provides better accuracy and loss value in training, validation and testing compared to other two optimizers and split ratio. The accuracy and loss obtained per each epoch during training of the CNN model using SGD optimizer with 80:20 split ratios is depicted in **Fig. 4** The model demonstrates an accuracy of 0.977 for training, 0.987 for validation and 0.980 for testing while utilizing the SGD optimizer.

Further fixing the optimizer and split ratio, the performance of the model observed with six different image resolutions is presented in **Table 2**. The 100 x 100 pixel image size results in the best training accuracy and is the fastest to compile and validate. The image size of 227x227 shows a minor reduction is the training accuracy but has comparable validation and testing accuracy with the smaller 100 x 100 image, and is the second fastest to compile. While other resolutions show no major improvement or sometimes lesser than that of 100 x 100 image and 227 x 227 image, they also take a longer time for training the model. That makes 100 x 100 and 227 x 227 image sizes the best fit image resolutions for the said CNN model. However from **Fig. 2, it** is observed that 100 x 100 image shows drastic reduction is image quality when plotted. Therefore 227 x 227 image size was set for further testing as it has better image clarity than that of 100 x 100 image while being the second fastest to compile the model training, with an overall better performance with respect to training, validation and testing accuracy.

Finally the model is also compared with VGG-16 a cutting-edge deep learning model. The main features of the suggested simple CNN's architecture with VGG-16 are shown in **Table 3**. The VGG-16 model has a total of 16 layers, that include one input layer, thirteen convolution layers, one flatten and one fully connected dense layer. It was trained upon the 'imagenet' dataset consisting millions of images with thousands of classes. Input image size needs to set at 250 x 250 pixels for the VGG-16 network. Initially the performance of the pre-trained VGG-16 for signature forgery detection is accessed.

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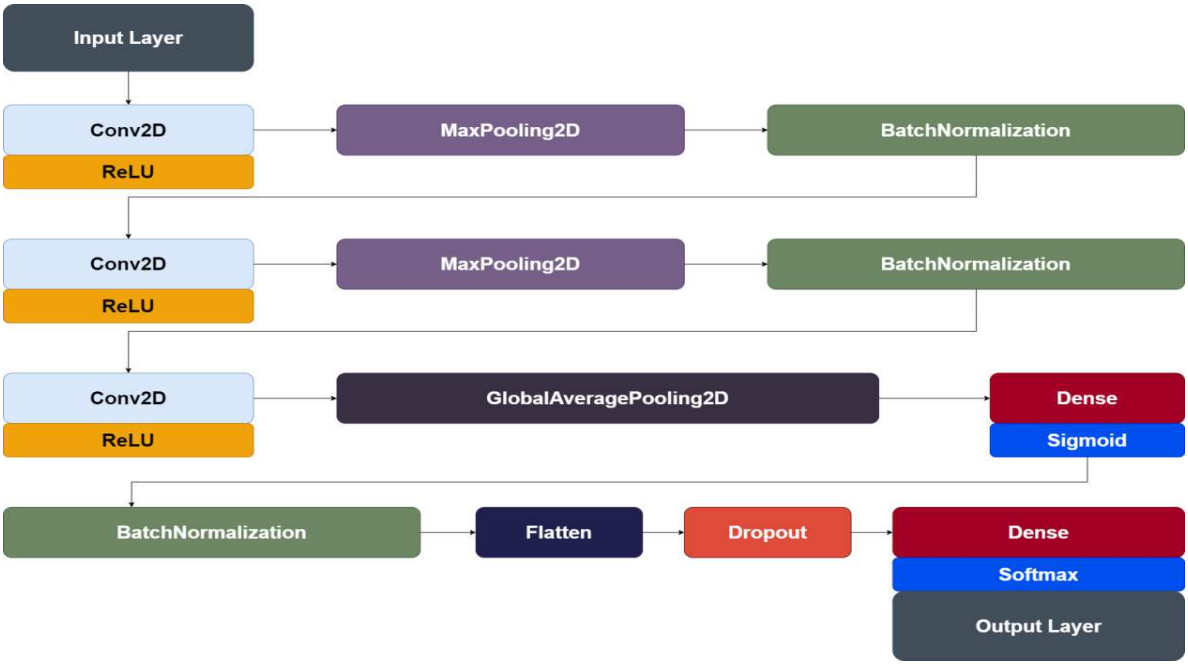


Fig.1 Architecture of the proposed CNN for SFDS

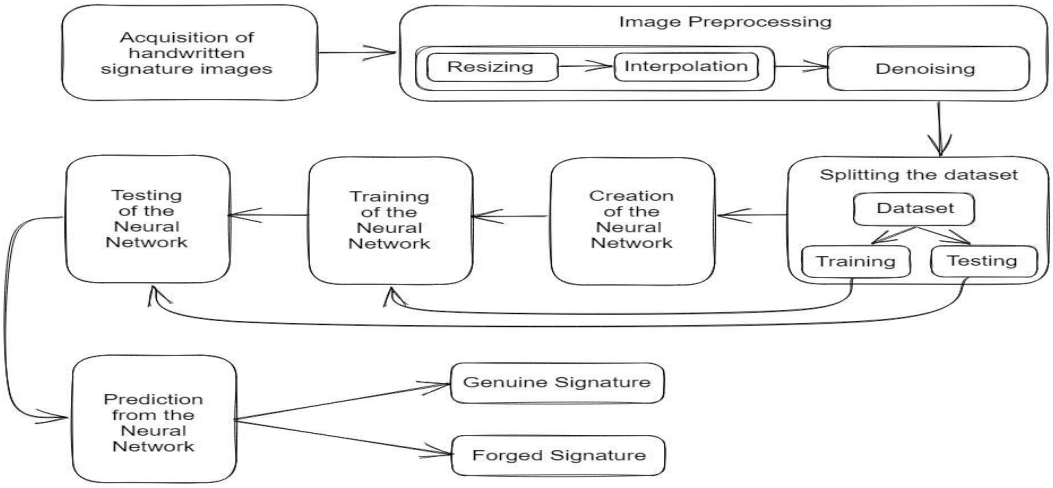


Fig. 2 CNN based SFDS framework



Fig. 3 Resized (in six different resolutions) and filtered pre-processed signature image along with the original one

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Table 1. Performance of the proposed CNN under different optimizers

SPLIT RATIO	Training Accuracy / Loss			Validation Accuracy / Loss			Test Accuracy / Loss		
	ADAM	RMSprop	SGD	ADAM	RMSprop	SGD	ADAM	RMSprop	SGD
70:30	0.997 / 0.003	0.990 / 0.024	0.977 / 0.052	0.889 / 0.294	0.47 / 20.996	0.981 / 0.039	0.886 / 0.337	0.526 / 19.443	0.972 / 0.049
80:20	0.998 / 0.003	0.995 / 0.011	0.979 / 0.053	0.459 / 26.199	0.45 / 31.65	**0.988 / 0.035**	0.509 / 23.787	0.509 / 28.713	**0.981 / 0.044**
85:15	0.999 / 0.001	0.506 / 0.693	0.977 / 0.051	0.459 / 25.723	0.459 / 27.622	0.987 / 0.038	0.507 / 23.456	0.507 / 25.185	0.980 / 0.047

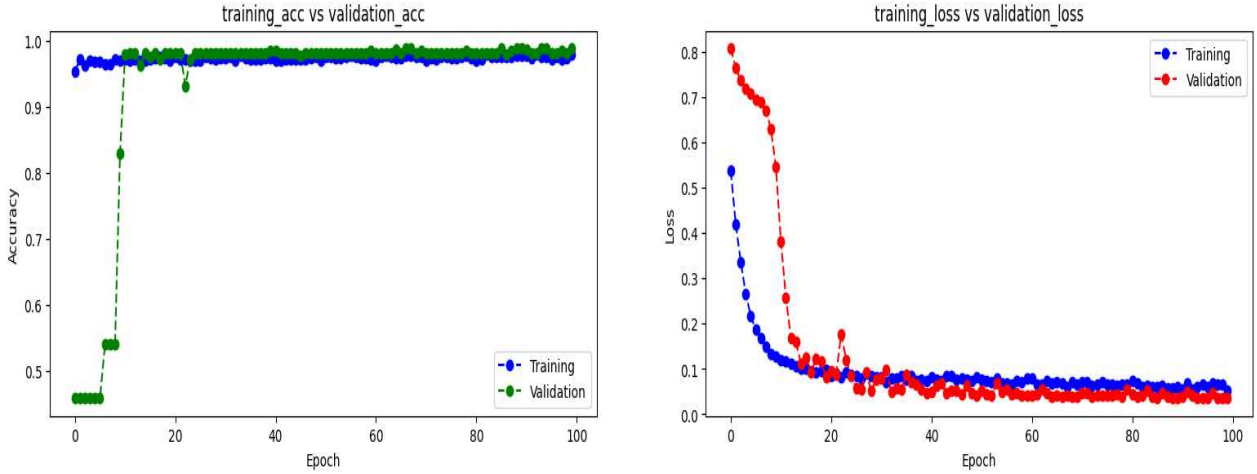


Fig. 4 Epochs vs. Training and Validation Accuracy / Loss

Table 2. Performance of the proposed CNN model with respect to various image sizes

IMAGE SIZE (in pixels)	TRAINING ACC.	VALIDATION ACC.	TESTING ACC.
100 x 100	0.981	0.988	0.985
227 x 227	0.979	0.988	0.981
250 x 250	0.975	0.984	0.985
250 x 100	0.972	0.981	0.981
350 x 100	0.977	0.988	0.983
400 x 150	0.974	0.981	0.978

Then the VGG-16 model is trained using SGD optimizer with learning rate 0.001 with the 80% of CEDAR dataset and its performance is contrasted with the pre-trained VGG-16 and the proposed simple CNN based SFDS model. **Table 4** shows the accuracy comparison of pre-trained and trainable VGG-16 models with the suggested one. Finally the performance of the three CNN models in term of accuracy, precision, recall and F1-score for the test set is depicted in **Fig. 5**.

The pre-trained VGG-16 clearly shows poor performance for the said classification task of handwritten signature forgery detection, taking a much longer time compared to the given 7-layer CNN model. When the layers of the VGG-16 model are unfrozen and it is made trainable, it shows better performance than the pre-trained VGG-16, but poor performance compared to the proposed CNN. The custom CNN based SFDS is able to achieve an accuracy of 0.981, precision of 1, recall of 0.967 and F1-score of 0.983, which is comparatively better than the other two models. This is due to the VGG-16 model not utilizing any batch normalisation filter in its convolution layers, while having no dropout layer and only using a single

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dense layer. The proposed 7-layer CNN model while having drastically less number of parameters in the network and having a significant size difference performs marginally better than the VGG-16 model.

IV. CONCLUSION

In this study, performance of three DNN based SFDS models such as the proposed simple 7 layer CNN, pre-trained VGG-16, trainable VGG-16 are observed for detecting forged and genuine cases from signature images. By training the model over the popular CEDAR dataset and analysing its results over validation and test set, it is evident that the suggested CNN based SFDS model having a simple architecture with fewer layers able to detect the forgery images from the image pool with a greater accuracy compared to the other two models. Further observing the model performance with three optimizers and three split ratios, it is clear that the SGD optimizer with split ration of 80:20 is best suitable for the proposed CNN model yielding the best training, validation and testing accuracy. Despite the remarkable accuracy generated by the proposed model, further improvements can be made by using data augmentation to generate a huge synthetic dataset to enhance the training of the model. In future, different hyper parameters can be tuned and adjusted by optimization techniques for further refinement of the CNN model.

Table 3. Architecture comparison of the proposed CNN model with VGG-16

Configuration	Proposed 7-Layer CNN Model	VGG-16
Convolution Layers	3	13
Dense Layers	2	1
Flatten Layers	1	1
Dropout Layers	1	0
No. of Pooling	3	5
Input Image Size	227 x 227	250 x 250
Trainable Parameters	110,466	50,178
Non Trainable Parameters	448	14,714,688
Total Parameters	110,914	14,764,866
Model Size	475KB	533MB

Table 4. Accuracy comparison of the proposed CNN model with VGG-16

Accuracy	Pretrained VGG-16	Trainable VGG-16	7-Layer CNN
Training	0.802	0.997	0.979
Validation	0.769	0.959	0.988
Testing	0.817	0.962	0.981

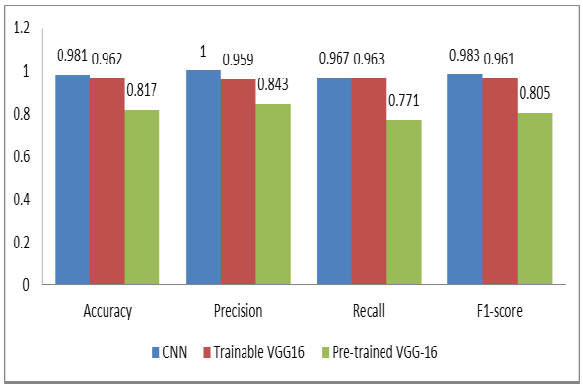


Fig. 5 Performance comparison of the three CNN based SFDS models

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